

# A More Credible Approach to Parallel Trends

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This paper proposes tools for robust inference in difference-in-differences and event-study designs where the parallel trends assumption may be violated. Instead of requiring that parallel trends holds exactly, we impose restrictions on how different the post-treatment violations of parallel trends can be from the pre-treatment differences in trends (“pre-trends”). The causal parameter of interest is partially identified under these restrictions. We introduce two approaches that guarantee uniformly valid inference under the imposed restrictions, and we derive novel results showing that they have desirable power properties in our context. We illustrate how economic knowledge can inform the restrictions on the possible violations of parallel trends in two economic applications. We also highlight how our approach can be used to conduct sensitivity analyses showing what causal conclusions can be drawn under various restrictions on the possible violations of the parallel trends assumption.

**Key words:** Difference-in-differences, Event-study, Parallel trends, Robust inference, Sensitivity analysis, Partial identification

**JEL codes:** C1

## 1. INTRODUCTION

Researchers using difference-in-differences (DiD) and related methods are often unsure about the validity of the parallel trends assumption needed for point identification of the causal parameter of interest. It has therefore become common practice to assess the plausibility of the parallel trends assumption by testing for pre-treatment differences in trends (“pre-trends”). Although pre-trends tests are intuitive, recent research has shown that they may suffer from low power (Freyaldenhoven *et al.*, 2019; Bilinski and Hatfield, 2020; Kahn-Lang and Lang, 2020; Roth, 2022), and that conditioning the analysis on passing pre-trends tests introduces statistical issues related to pre-testing (Roth, 2022). How then should researchers proceed when they are unsure about the validity of the parallel trends assumption?

This paper proposes methods for robust inference and sensitivity analysis in empirical settings where the parallel trends assumption may not hold. Building on work by Manski and Pepper (2018), we show that the causal parameter of interest can be (partially) identified under

pre-treatment violation of parallel trends. As a sensitivity analysis, the researcher might also report confidence sets that allow the maximum post-treatment violation of parallel trends to be up to  $\bar{M}$  times larger than the maximum pre-treatment violation for different values of  $\bar{M}$ . The proposed sensitivity analysis thus follows the “top-down” approach described by Tamer (2010), in which the researcher reports what can be learned under a sequence of progressively weaker assumptions. Performing such sensitivity analyses makes clear what must be assumed about the possible differences in trends in order to draw specific causal conclusions, in line with the partial identification paradigm initiated by Charles Manski and co-authors in a variety of contexts (Manski, 1989, 1990, 2003, 2008). We provide the HonestDiD R and Stata packages to implement our methods.<sup>1</sup> We illustrate our recommended approach with applications to two recently published papers, in which we show how the choice of the restrictions  $\Delta$  can be tailored to the economic context.

### 1.1. Related literature

The approach in this paper builds on the foundational partial identification analysis for DiD in Manski and Pepper (2018). Manski and Pepper consider identification under researcher-specified bounds on the magnitude of  $\delta_{post}$  (what they call “bounded DiD variation”), and calibrate these bounds using the maximal pre-treatment violation of parallel trends in their empirical application on the effects of right-to-carry gun laws.<sup>2</sup> One of our leading classes of restrictions,  $\Delta^{RM}$ , formalizes this calibration approach by bounding the magnitude of post-treatment violations of parallel trends by  $\bar{M}$  times the maximal pre-treatment violation. Our framework also allows for many other intuitive restrictions—such as bounds on how far  $\delta$  can deviate from linearity—and it can be applied to a variety of DiD estimators, including recent proposals for settings with staggered treatment timing. Most importantly, while Manski and Pepper (2018) provide a framework for identification, we provide inference methods to construct uniformly valid confidence sets for the treatment effect of interest. This allows applied researchers to account for statistical uncertainty in their analyses, which can be important since event-study coefficients are often imprecisely estimated in practice.

Several other recent papers consider various relaxations of the parallel trends assumption. Keele *et al.* (2019) develop techniques for testing the sensitivity of DiD designs to violations of the parallel trends assumption, but they do not incorporate information from the observed pre-trends in their sensitivity analysis. Empirical researchers commonly adjust for the extrapolation of a linear trend from the pre-treatment periods when there are concerns about violations of the parallel trends assumption, which is valid if the difference in trends is exactly linear (*e.g.* Bhuller *et al.*, 2013; Dobkin *et al.*, 2018; Goodman-Bacon, 2018, 2021). Our methods nest this approach as a special case, but allow for valid inference under less restrictive assumptions about the class of possible differences in trends (such as when  $\delta$  is only approximately linear). Freyaldenhoven *et al.* (2019) propose a method that allows for violations of the parallel trends assumption but requires an additional covariate that is affected by the same confounding factors as the outcome but not by the treatment of interest. Ye *et al.* (2020) consider partial identification of treatment effects when there exist two control groups whose outcomes have a bracketing relationship with

1. The latest version of the R and Stata packages are respectively available at <https://github.com/asheshrambachan/HonestDiD> and <https://github.com/mcaceresb/stata-honestdid/>.

2. Manski and Pepper (2018) also consider “bounded time” and “bounded state” restrictions that bound how much the mean of  $Y(0)$  can differ either across treatment groups or within-groups over time. Such restrictions could also be incorporated into our framework by augmenting the vector  $\hat{\beta}$  to include group-specific sample averages.